DATA ANALYSIS | SAMPLE SUPERSTORE

The Superstore Data Looks Healthy Until You Look at Discounts

Nine thousand order lines, one uncomfortable pattern

$2.3M	12.5%	96.8%
in sales, 2014 to 2017	overall profit margin	of lines above 20% off lost money

On paper, this business looks fine. The Sample Superstore data shows about **$2.3 million in sales** over four years and roughly **$286,000 in profit**. That is a 12.5% margin. Not amazing, not scary.

But one number kept pulling me back. When a discount went above 20%, **96.8% of those order lines lost money.** Not most of them. Almost all of them.

This article is about how I got to that number, what I trust about it, and what I don't.

First, the Boring Part: The Data

The file has 9,994 rows and 21 columns, covering January 2014 to December 2017. No missing values, no duplicates. Easy start, right? Not quite.

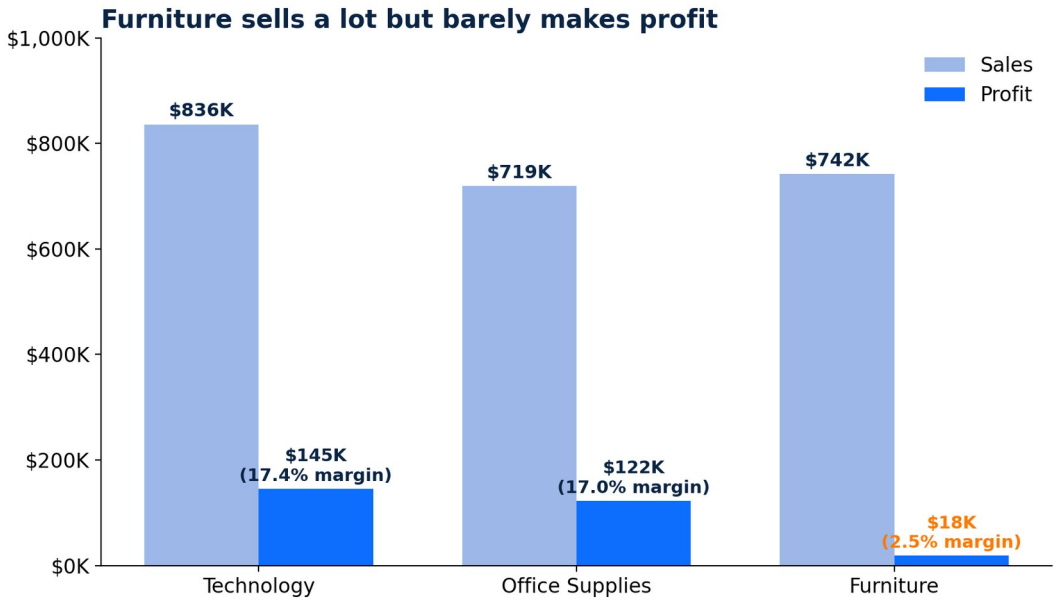
It would not open as UTF-8, so I had to read it with latin-1 encoding. Postal codes were stored as numbers, which means New England zip codes lost their leading zero (the smallest one in the file shows up as 1040). And the one that matters most: **each row is an order line, not an order.** There are 9,994 rows but only 5,009 unique orders. If you count rows and call them orders, you are off by about half.

None of this is dramatic. But it is exactly the kind of thing that quietly breaks a number later.

Sales Is the Wrong Place to Start

My first instinct was to rank categories by sales, because that is what most dashboards do. Technology came out on top at $836K, then Furniture at $742K, then Office Supplies at $719K.

Then I added profit, and the picture changed.

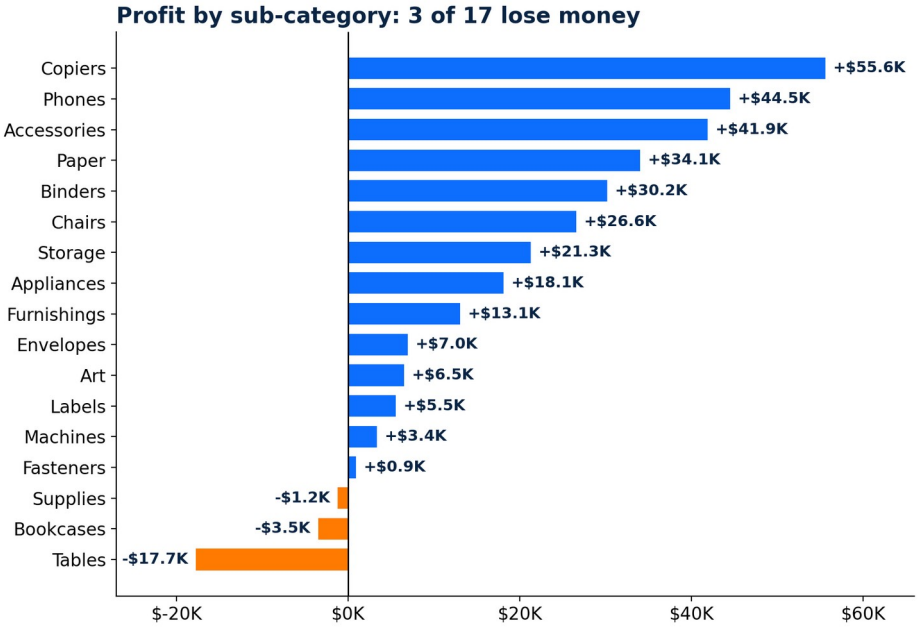


Sales vs profit by category

Furniture sells almost exactly as much as Office Supplies but keeps about **15% of the profit**: $18K against $122K. That is a 2.5% margin, compared with about 17% for the other two. Same effort at the till, very different result.

Where the Losses Hide

Digging one level down, 3 of the 17 sub-categories lose money: **Tables (−$17.7K), Bookcases (−$3.5K) and Supplies (−$1.2K).** Tables is the real problem here. The other two are small.



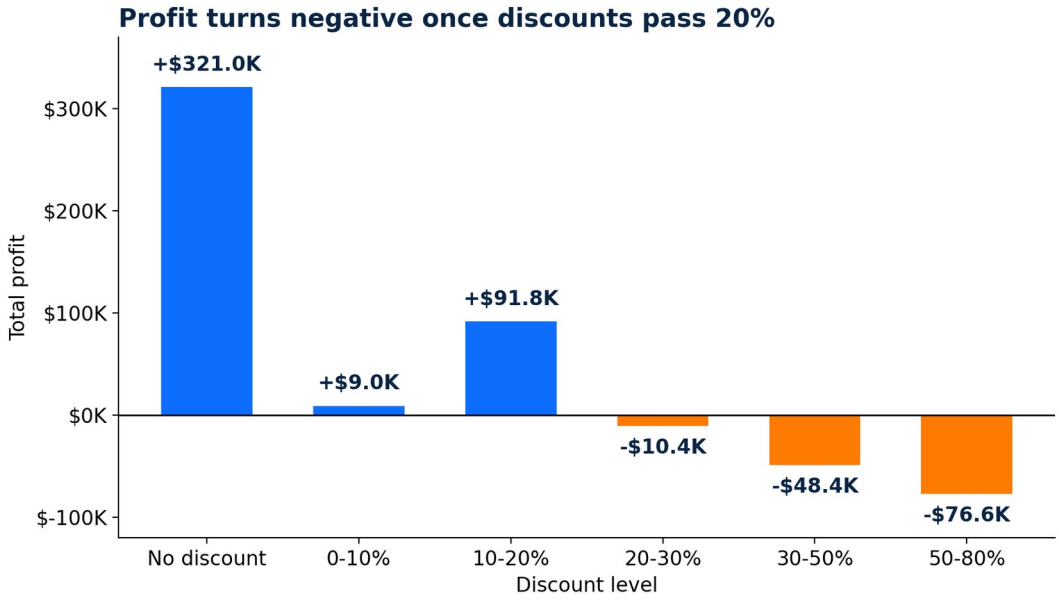
Profit by sub-category

Machines caught my eye too. It brought in $189K in sales and only $3.4K in profit. On the other side, Copiers had just 68 order lines but made $55.6K at a 37% margin, the best of any sub-category.

A warning on that last one: the three single most profitable lines in the whole file are the same copier model. The top 10 lines alone make up 15.6% of all profit. So a lot of the “good news” rests on a handful of big orders.

The Discount Cliff

Now the part that bothered me. I grouped every order line by its discount and added up the profit in each group.



Total profit by discount level

Lines with no discount made **$321K**. Lines with 10 to 20% off still made **$92K**. Then it flips: 20 to 30% off lost $10K, 30 to 50% lost $48K, and 50 to 80% lost $77K.

Put another way: the 1,393 lines discounted by more than 20% (about 14% of all lines) lost **$135K together**. The other 8,601 lines made **$422K**. And every single line that lost money had some discount attached, though plenty of discounted lines were fine.

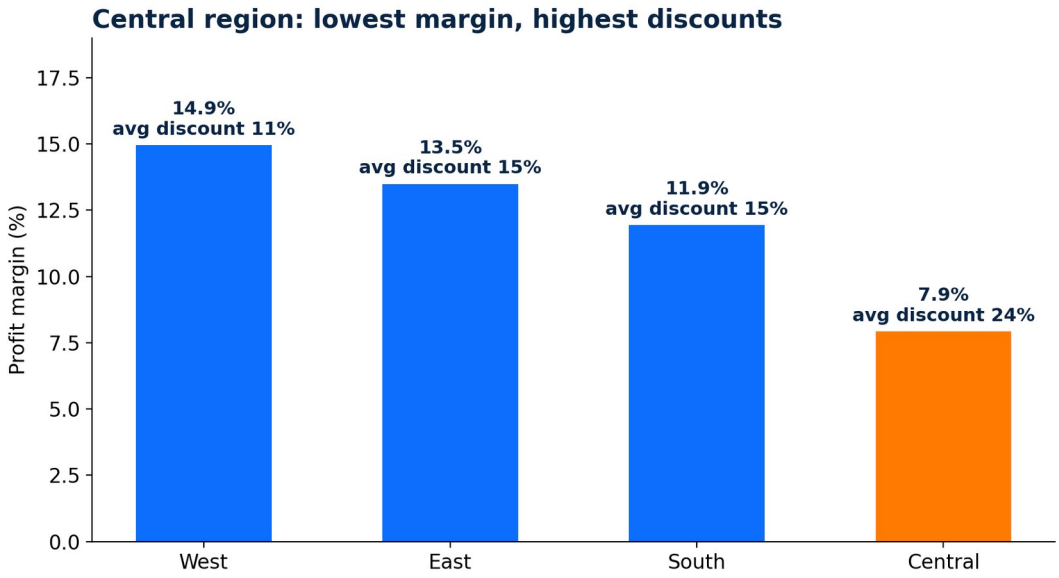
> **Careful: this is a pattern, not proof that discounts cause the losses.**

Here is why I am hesitant to say “discounts cause losses.” The overall correlation between discount and profit is only −0.22, which is weak. The damage is not a smooth slope. It is concentrated in specific products. Tables average a 26% discount, Machines 31%, and Bookcases 21%, while Paper sits near 8%. Maybe those products were priced thin to begin with and the discount finished them off. Maybe the discounts were the only way to sell them at all.

And Supplies loses money with an average discount of only about 8%, so discounts are clearly not the whole story.

Regions and States

The same pattern shows up on the map. The Central region has the lowest margin, **7.9%**, and the highest average discount, **24%**. The West sits at 14.9% margin with an average discount of 11%.



Profit margin by region

At state level, the five worst performers are Texas (−$25.7K), Ohio (−$17.0K), Pennsylvania (−$15.6K), Illinois (−$12.6K) and North Carolina (−$7.5K), about **$78K in losses combined**. Together they average a **35% discount**, against roughly 8% in every other state. California and New York, by contrast, bring in $76K and $74K.

One more thing worth saying: the business did grow. Sales went from $484K in 2014 to $733K in 2017, and the margin improved from 10.2% to 12.7%. A growing, improving business is exactly the kind that never goes looking for a discount problem.

What I Would Suggest (and What I Wouldn't)

If I were handing this to a manager, I would not say "stop all discounts." The data cannot tell me how many of those discounted orders would have happened without the discount. Some probably would not have.

What I would suggest is a **test, not a rule**: cap discounts at 20% for Tables and Machines, or in a few weak states, and compare margin against a similar group that keeps the old policy. That gives you an actual answer instead of my guess.

What This Dataset Cannot Tell Us

It is a **sample practice dataset**, so none of this describes a real company. It also has no cost data beyond profit, no reason for any discount, and no information on what customers did next. Treat the findings as a good exercise in careful reading, not a business verdict.

What I Took From It

Sales tells you how busy you are. Profit tells you whether being busy was worth it. A total that looks healthy can hide a corner of the business that is quietly losing money, and you only find it if you keep splitting the data until something looks wrong.

I also learned to slow down with causes. It is tempting to write a neat headline like “discounts destroy profit.” The honest version is longer and less catchy, but it is the one I can defend.

If you have worked on retail data: **what is the first thing you check?** And would you have looked at discounts before regions? Tell me in the comments.

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